

# Arpit Kumar Singh

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## SUMMARY

Engineering student with hands-on experience in designing reliable, scalable web systems and strong grounding in data structures, algorithms, and system design fundamentals. Passionate about writing clean code, learning new technologies, and building products that prioritize reliability, performance, and user impact.

## EDUCATION

**Netaji Subhas University of Technology (NSUT)** New Delhi, India  
B.Tech in Mechanical Engineering 2023 – 2027

**Indian Institute of Technology Madras (IIT Madras)** Tamil Nadu, India  
B.Sc. in Data Science and Applications 2024 – 2027

## WORK EXPERIENCE

**CAE Design Intern** May 2026 - Jun 2026

- Developed detailed 3D CAD models and performed FEA simulations including static structural, modal, thermal, fatigue, and dynamic analysis using ANSYS and SolidWorks.
- Validated simulation results through stress, deformation, strain, temperature, and safety factor evaluation to ensure design reliability and performance.

## PROJECTS

**StayIn.com** MongoDB, JWT Auth, REST APIs

- Designed a **full-stack** booking platform inspired by Airbnb, focusing on reliability and data-consistency across listings, search, booking workflows, and user dashboards ( **90%** feature parity).
- Implemented a booking availability engine with concurrency-safe validation and transaction checks to prevent race-condition conflicts, enabling **99%** successful bookings across **200+ tests**.
- Adopted the **MVC (Model–View–Controller)** architecture to structure the application, improving code modularity, maintainability, and separation of concerns across APIs, business logic, and UI components.

**E-Commerce Platform** Node.js, Express, MongoDB

- Re-designed database schema and query structure using normalization and indexing strategies, achieving **20% lower query latency** while preserving correctness for analytics and dashboards.
- Implemented **real-time cart and wishlist** activity logging to study retention trends and data-driven product decisions.

**Credit Card Fraud Detection Using Machine Learning** Python, Scikit-learn, Pandas, NumPy

- Developed a fraud detection system using **Logistic Regression** to classify legitimate and fraudulent credit card transactions.
- Performed data cleaning, feature analysis, and exploratory data analysis (EDA), addressing severe class imbalance through **undersampling**.
- Trained and evaluated the model on a balanced dataset of **37,721 transactions** with **30 predictive features**.
- Achieved **97.48% training accuracy** and **98.32% testing accuracy**, ensuring reliable fraud detection performance.
- Built an end-to-end prediction pipeline for real-time transaction classification and model evaluation.

## POSITIONS OF RESPONSIBILITY

**Joint Secretary — IEEE NSUT** Dec 2023 – Present

- Coordinated **TENSYP'24** — Asia-Pacific conference for researchers, improving cross-team communication and execution.
- Conducted **Web Development and DSA bootcamps**, mentoring students on coding practices and project-based learning.
- Directed a **40+ member team**, streamlining workflows and supporting junior contributors.

**Sponsorship Coordinator — Moksha Innovation 2024** Jan 2024 – Mar 2024

- Collaborated with brands** including SPARKS, Decathlon, Vision IAS, and Storia to drive sponsor engagement.
- Facilitated sponsorship negotiations and MOU workflows, achieving **6K+ app downloads** for the co-title sponsor.

## ACHIEVEMENTS

- Hackathon** — Developed and deployed a full-stack product in a **24-hour hackathon**, demonstrating rapid prototyping and collaborative problem solving.
- Career Essentials in Software Development — **Microsoft + LinkedIn**
- Elite Certificate** — Design Thinking (IITM), Soft Skills & Personality (IITM)